

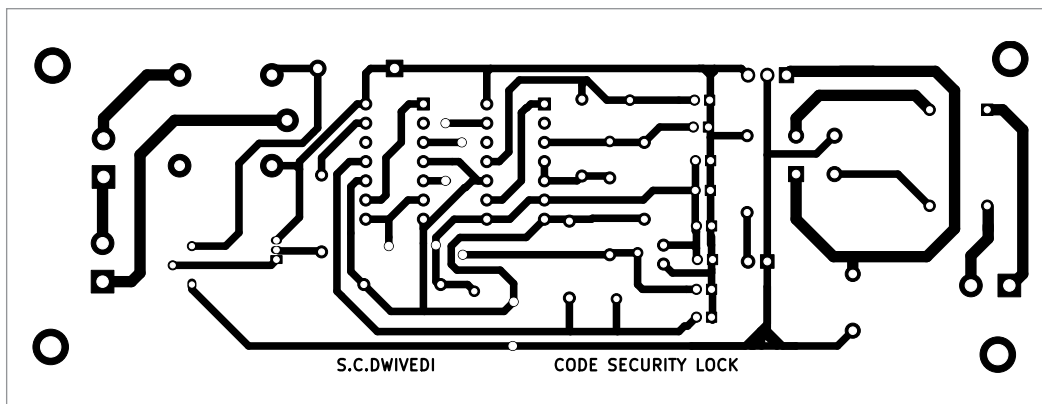


Fig. 3: Actual-size PCB layout of the circuit

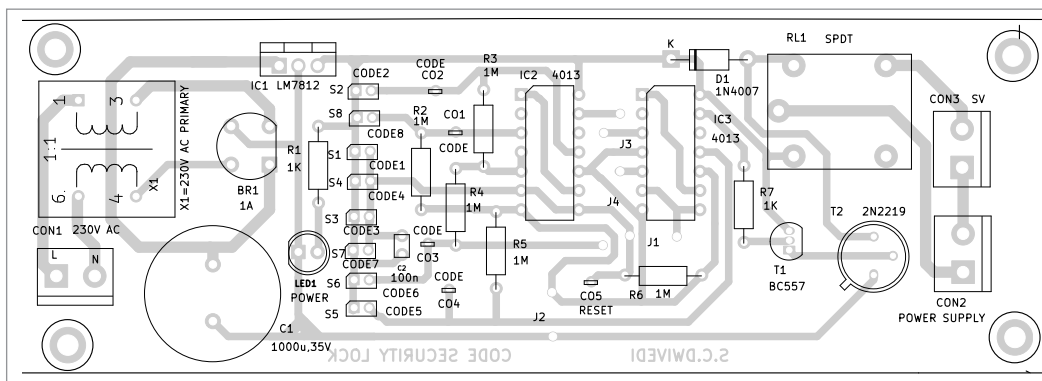


Fig. 4: Component side of the PCB

the respective switch S1 to CO1, S7 to CO2, S5 to CO3, and S6 to CO4, and connect the rest of the switches, such as S2, S3, S4, and S8, to the reset point CO5. However, in this

PARTS LIST

Semiconductors:

- IC1 - LM7812, 5V voltage regulator
- IC2 (IC2A-IC2C), IC3 (IC3A-IC3C) - 4013 D-flip-flop
- D1 - 1N4007 rectifier diode
- LED1 - 5mm LED
- T1 - BC557 pnp transistor
- T2 - 2N2219 npn transistor

Resistors (all 1/4-watt, $\pm 5\%$ carbon):

- R1, R7 - 1-kilo-ohm
- R2-R6 - 1-mega-ohm

Capacitors:

- C1 - 1000 μ F, 35V electrolytic
- C2 - 100nF ceramic disk

Miscellaneous:

- CON1-CON3 - 2-pin connector
- SV - 12V solenoid valve
- RL1 - 12V, 1C/O (SPDT) relay
- X1 - 230V AC primary to 15V, 500mA secondary transformer

project, the set code is 8265. Therefore, the respective switches S8, S2, S6, and S5 are connected to CO1 through CO4, and the remaining switches (S1, S3, S4, S7) are connected to reset point CO5, as shown in the schematic (Fig. 2).

If the correct clocking sequence is followed, a low level (GND) appears at Q2 output (pin 13) of IC3, which energises the relay through relay driver transistors T1 and T2. As a result, the solenoid activates and the lock opens. The reset keys are connected to the set pins 6 through 8 of IC2 and IC3. The power-on-reset capacitor C2 is used to stabilise the pulses during power-on of the circuit.

Construction and testing

An actual-size, single-side PCB layout for the lock is shown in

Fig. 3 and its component layout is shown in Fig. 4. After assembling the circuit on PCB, enclose it in a suitable box. Fix LED1 and CON1 on the front side of the box, and fix connectors CON2 and CON3 on the rear side of the box. Use CON2 for connecting the power supply of the solenoid valve and CON3 for connecting the solenoid valve.

Installation

First, create a 4x4 keypad matrix. Connect the respective switches (based on the code lock) to CO1 through CO4 and

connect the remaining switches to the reset point CO5. Now, install this keypad at the entrance door and connect the solenoid valve to the circuit. Your circuit is now ready to use.

You can also install this simple code lock circuit in your car so that it starts only when the correct code sequence is entered via the keypad. The code lock design can also be used in various other applications.

Bonus. You can watch the video of the tutorial of this DIY circuit idea at <https://www.electronicsforu.com/videos-slideshows/live-diy-code-security-lock> **EFY**

S.C. Dwivedi is an electronics enthusiast and circuit designer at EFY